A Course-Based Project Report on

CLIENT INTELLIGENCE CENTER

Submitted to the

Department of CSE-(CyS, DS) and AI&DS

in partial fulfillment of the requirements for the completion of the course

Big Data Computing LABORATORY (22PC2DS303)

BACHELOR OF TECHNOLOGY

in

CSE- DATA SCIENCE

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CERTIFICATE

This is to certify that the project report entitled **"CLIENT INTELLIGENCE CENTER"** is bonafide work done under our supervision and is being submitted by **K.HANISH(22071A6786), M.LAXMIPATHI BALAJI (22071A6796) ,V.KARTHIKEYA(22071A67C7), Y.MITHIL(22071A67C8)** in partial fulfillment for the award of the degree of **Bachelor of Technology in Computer Science and Engineering and Data Science**, of the VNRVJIET, Hyderabad during the academic year 2025-2026.

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DECLARATION

e declare that the course-based project work entitled "**CLIENT INTELLIGENCE CENTER**" submitted the Department of **Computer Science and Engineering and Data Science**, Vallurupalli Nageswara Rao gnana Jyothi Institute of Engineering and Technology, Hyderabad, in partial fulfillment of the requirement the award of the degree of **Bachelor of Technology in Artificial Intelligence & Data Science** is a bonafide ord of our work carried out under the supervision of **Mrs.P.Meenakshi (PhD), Assistant Professor, partment of CSE- (CyS, DS) and AI&DS, VNRVJIET.** Also, we declare that the matter embodied in this sis has not been submitted by us in full or in any part thereof for the award of any degree/diploma of any er institution or university previously.

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ABSTRACT

'he **Client Intelligence Center** project leverages the Hadoop ecosystem to analyze user ehavior data, uncovering patterns and trends critical for enhancing customer engagement and xperience. Utilizing Hadoop Distributed File System (HDFS) for data storage, MapReduce for ata processing, Hive for data querying, and Pig for data transformation, this project enables fficient handling and analysis of vast amounts of user data. Sqoop is implemented to facilitate eamless integration between traditional SQL databases and Hadoop, ensuring comprehensive ata availability for analysis. This project aims to provide actionable insights into customer ehavior, driving informed business decisions.

INTRODUCTION

In the digital age, understanding user behavior is paramount for businesses aiming to stay ompetitive. The sheer volume of data generated by user interactions necessitates robust tools nd frameworks for efficient analysis. The Hadoop ecosystem, with its distributed storage and rocessing capabilities, presents an ideal solution for managing and analyzing large datasets. In his project, we will create a comprehensive analysis pipeline for user data using the Hadoop cosystem. We will store and process large datasets containing user interactions, then analyze nd extract meaningful insights. The project will involve setting up a Hadoop Distributed File ystem (HDFS) for data storage, processing the data with MapReduce, querying with Hive, and cripting data transformations with Pig. We will also use Sqoop to facilitate data transfer etween SQL databases and Hadoop.

METHODOLOGY

Data Collection and Storage:

- **HDFS**: User interaction data is ingested and stored in the Hadoop Distributed File System (HDFS). Below is an example of how data might be stored in HDFS using Hadoop

Commands:

```
hadoop fs -put /local/path/to/user_interaction_data /user/hadoop/data/
```

Data Integration:

- **Sqoop**: Data from existing SQL databases is imported into HDFS using Sqoop. Here's a snippet demonstrating Sqoop usage to import data from MySQL:

```
sqoop import --connect jdbc:mysql://mysql_server/blogspot_db --username user --password pass --table user_interactions --target-dir /user/hadoop/data/
```

Data Processing:

- **MapReduce**: MapReduce jobs are implemented for processing large datasets in a distributed manner. Here's a simplified example of a MapReduce program to count user interactions:

```
public class UserInteractionCount {
    public static void main(String[] args) throws Exception {
        Job job = Job.getInstance(new Configuration(), "User Interaction Count");
        job.setJarByClass(UserInteractionCount.class);
        FileInputFormat.addInputPath(job, new Path("/user/hadoop/data/user_interactions"));
        FileOutputFormat.setOutputPath(job, new Path("/user/hadoop/output/interaction_count"));
        job.setMapperClass(InteractionMapper.class);
        job.setReducerClass(InteractionReducer.class);
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

Data Querying:

- **Hive**: Hive queries are used for data querying and analysis. Here's an example of a Hive query to find the average user interaction duration:

```
SELECT user_id, AVG(duration) AS avg_duration
FROM user_interactions
GROUP BY user_id;
```

Data Transformation:

- **Pig**: Pig scripts are utilized for data transformation tasks. Below is an example of a Pig script to filter and transform user interaction data:

```
user_interactions = LOAD '/user/hadoop/data/user_interactions' USING PigStorage(',') AS (user_id:int, action:chararray, timestamp:long);
filtered_interactions = FILTER user_interactions BY action == 'click';
grouped_by_user = GROUP filtered_interactions BY user_id;
user_interaction_count = FOREACH grouped_by_user GENERATE group AS user_id, COUNT(filtered_interactions) AS click_count;
STORE user_interaction_count INTO '/user/hadoop/output/user_interaction_count' USING PigStorage(',');
```

Data Analysis and Visualization:

- Insights extracted through Hive queries and Pig scripts can be visualized using tools like Tableau or custom dashboards. Visualization aids in interpreting user behavior patterns effectively.

TEST CASES/OUTPUT

Test Case-1 :

```
1,Alice,30,Female,150.00,2024-01-01
2,Bob,25,Male,200.50,2024-02-15
3,Charlie,35,Male,300.75,2024-03-20
```

Test Case-2 :

```
customer_data_hive
```

Test Case 3:

```
1  Alice    30  Female  150.00  2024-01-01
2  Bob      25  Male    200.50  2024-02-15
3  Charlie  35  Male    300.75  2024-03-20
```

Test Case 4:

```
Female  150.00
Male    250.625
```

RESULTS

Import customer data from MySQL to HDFS.

```bash
sqoop import \
  --connect jdbc:mysql://localhost:3306/customer_db \
  --username your_username \
  --password your_password \
  --table customer_data \
  --target-dir /user/hadoop/customer_data \
  --m 1
```

Result: Data was successfully imported to HDFS.

Analyze customer data to derive insights such as average purchase amount by gender.

```sql
SELECT gender, AVG(purchase_amount) AS avg_purchase
FROM customer_data_hive
GROUP BY gender;
```

Result: Successfully derived insights from the customer data.

```plaintext
Female  150.00
Male    250.625
```

DISCUSSION

1. **Hadoop Distributed File System (HDFS):**
   - HDFS serves as the foundational storage layer in the project. It accommodates large volumes of user interaction data in a distributed manner, ensuring reliability and scalability. This capability is crucial as it allows the project to handle vast amounts of data generated from user interactions across various channels.
2. **MapReduce:**
   - MapReduce is employed for data processing tasks within the project. It facilitates parallel processing of large datasets across a distributed network of nodes. By breaking down complex data processing tasks into smaller manageable tasks (maps and reduces), MapReduce enables efficient computation and aggregation of user behavior patterns and trends.
3. **Hive:**
   - Hive is utilized for querying and analyzing the processed data stored in HDFS. Its SQL-like interface simplifies data exploration and extraction of insights from structured data sets. This capability is instrumental in enabling analysts and data scientists to derive actionable insights into customer behavior through ad-hoc queries and data summarization.
4. **Pig:**
   - Pig complements MapReduce by offering a high-level data flow language. It allows for the scripting of data transformations and manipulations that are executed on Hadoop. This flexibility is valuable in handling diverse data processing requirements, such as cleansing, transforming, and preparing data for analysis.
5. **Sqoop:**
   - Sqoop facilitates seamless data integration between traditional SQL databases and the Hadoop ecosystem. It supports the efficient import and export of data, ensuring that all relevant data sources are accessible for analysis within the Hadoop environment. This capability enables the project to leverage existing data assets from relational databases, enhancing the comprehensiveness of insights derived from user behavior data.

CONCLUSION

he **Customer Insights Hub** project successfully demonstrates the power of the Hadoop ecosystem in ıalyzing user behavior data. By leveraging HDFS for storage, MapReduce for processing, Hive for querying, ig for data transformation, and Sqoop for integration, the project provides a comprehensive solution for ctracting actionable insights from large datasets. These insights can drive informed business decisions, ltimately enhancing customer engagement and experience. The Hadoop ecosystem's scalability and flexibility ıake it an invaluable tool for businesses looking to harness the full potential of their data.

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